

Laboratory Report 2: Enzyme Kinetics

Due: Wednesday, October 28 at 11:59 PM Value: 100 points Format: Individual scientific report

Purpose

Evaluate how pH alters catalase activity and determine whether the evidence supports an optimum near neutral pH.

Your task

- Calculate mean oxygen-production rate and standard deviation for each pH from the replicate data.
- Create a figure showing means, individual replicates, and variability.
- Explain the curve using ionizable active-site groups and protein stability without claiming direct structural evidence.
- Address substrate concentration, temperature, foam capture, and enzyme preparation as potential sources of variation.
- Propose a follow-up that distinguishes reversible pH effects from irreversible denaturation.

Provided evidence or data

pH	Replicate 1	Replicate 2	Replicate 3
3	4.2	4.8	4.5
5	12.1	11.4	12.7
7	19.8	20.4	19.5
9	10.6	11.2	10.1
11	2.8	3.1	2.5

Submit

- 1,200-1,500 word scientific report PDF.
- Appendix with calculations and one sample mean/standard-deviation calculation.

Important note

Rates are milliliters of oxygen per minute and are simulated from the class protocol.

Grading criteria

Criterion	Pts	Full-credit evidence
Question and methods	15	Frames the investigation and describes the relevant procedure accurately.
Results	25	Calculations, figures, captions, and statistical summaries are correct and transparent.
Interpretation	30	Explains results using an appropriate cellular mechanism and addresses uncertainty.
Evaluation	20	Identifies limitations, controls, and a justified follow-up experiment.
Scientific form	10	Clear structure, accurate units, citations, and reproducible data presentation.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.